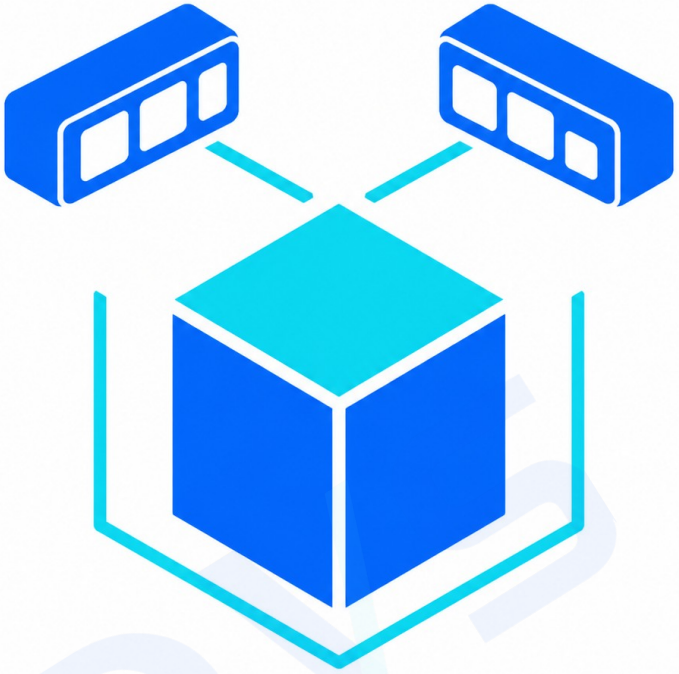
MRDVS
Mobile Robot Vision Expert



AW3 TorusMetric

Volume Measurement Solution User Manual

Measure object size and volume in 3D.

Zhejiang MRDVS Technology Co., Ltd.

V0.2 · 2026-10-28

Contents

1 Product Overview 3
1.1 Operating Prerequisites 3
1.2 First-Use Workflow 4
2 Before You Start 5
2.1 Recommended Operating Conditions 5
2.2 Conditions That May Prevent Detection or Measurement 6
2.3 Result Definitions and Acceptance Criteria 6
3 Complete the First Measurement 7
3.1 Main Interface 7
3.2 Multi-Camera Calibration 8
3.3 Ground Calibration 15
3.4 Set Detection Areas 17
3.5 Set Detection Parameters 19
3.6 Monitor and Verify Measurement Results 20
3.7 Start Daily Measurement 23
4 Functions and Parameters 24
4.1 Manage Detection Areas 24
4.2 Adjust Detection and Filtering Parameters 25
4.3 Merge Area Objects and Real Volume 26
4.4 RGB Camera, Point-Cloud Views, and Save Image 26
4.5 Calibration Parameters 28
5 Communication and Result Data 29
5.1 Configure and Verify Communication 29
5.2 Read Result Messages 29
6 Troubleshooting and Maintenance 31
6.1 System Issues (Streaming) 31
6.2 Algorithm Performance Issues 31
6.3 Export Troubleshooting Data 32
6.4 Maintenance and Revalidation 32
Revision History and Technical Support 33
Technical Support 33

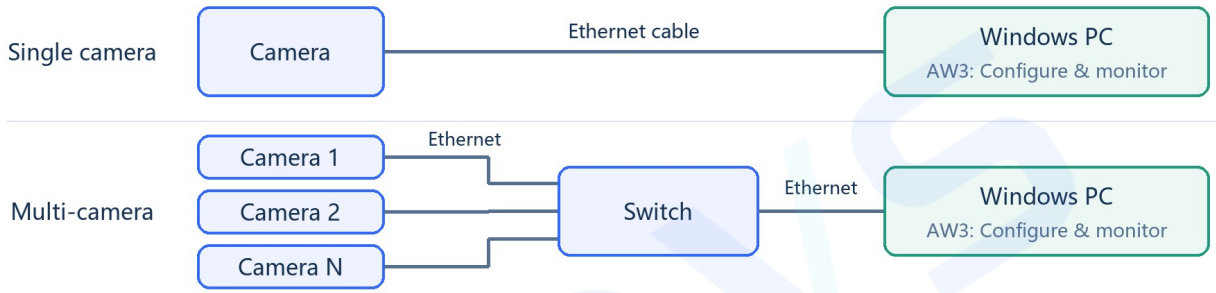
1 Product Overview

TorusMetric runs on AW3 and measures object length, width, height, and volume from 3D point clouds. It supports single- and multi-camera systems with up to five detection areas.

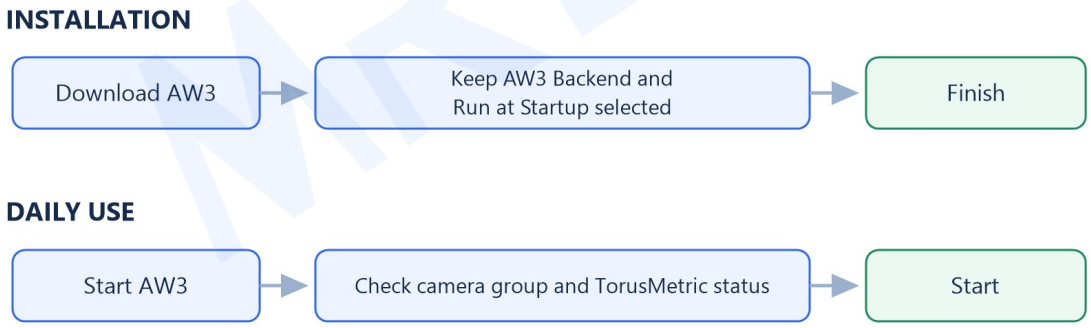
1.1 Operating Prerequisites

Hardware Installation and Connections

Use a Windows PC with AW3 installed for configuration and monitoring. Connect a single camera directly to the PC via Ethernet. For multiple cameras, connect each camera and the PC to the same switch using Ethernet cables. Multi-camera stitching requires a 12th-generation Intel Core i5 or above.



For TorusMetric, select [AW3 backend] and [Start the backend when Windows starts]. Select backend launch after installation. For daily use, confirm that the backend is running before starting AW3. See Chapter 4 of the AW3 User Manual.



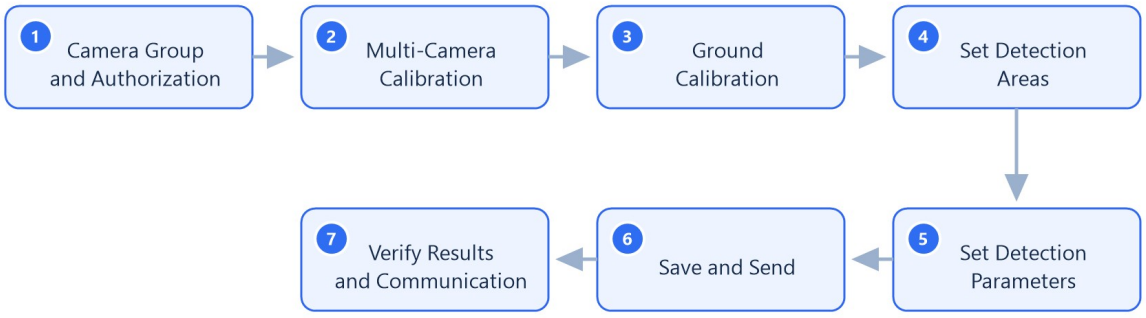
Keep the default installation options; the backend starts automatically.

Restart the current center Opening or configuring a camera in LanxincameraViewer may cause a central-service streaming error. If this occurs, open AW3 [Device Connection / Base Camera] > [Central Service] and select [Restart current center]. Wait for the connection and stream to recover, then verify measurement results.

Related manual **For all AW3 platform functions, including camera and device connections and software/firmware upgrades, refer to the AW3 User Manual.** This manual focuses on configuring and using TorusMetric.

1.2 First-Use Workflow

Complete the main workflow first, then use Chapter 4 for secondary functions and advanced parameters. Single-camera groups skip multi-camera stitching.



Single camera: skip Step 2.

First-use workflow

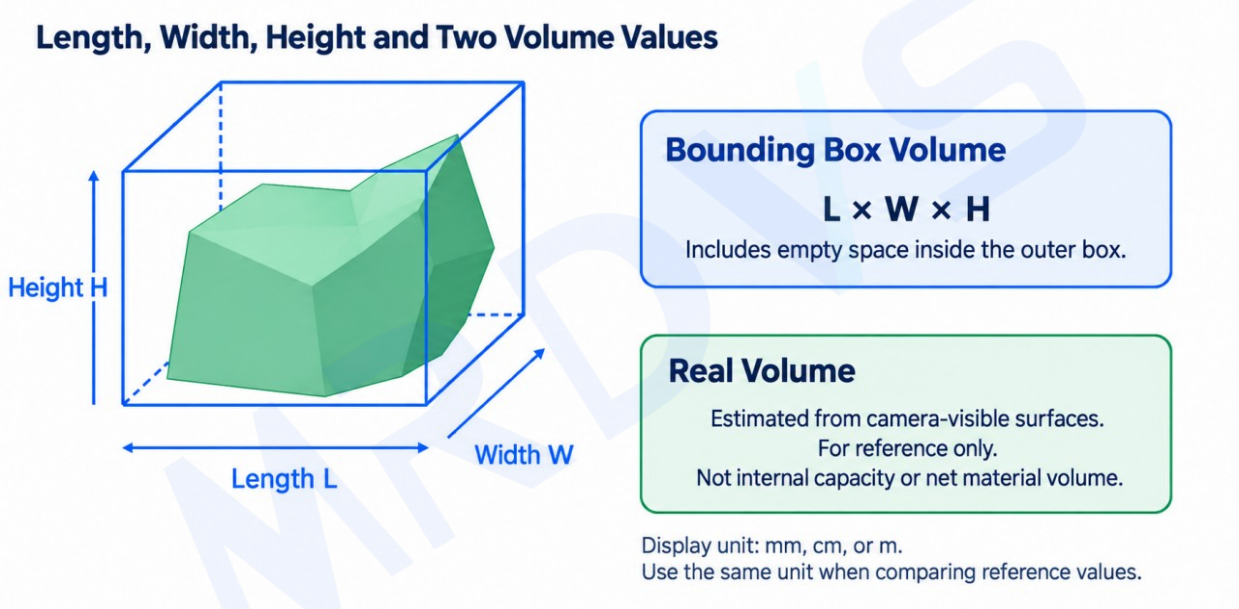
Calibrate first Complete the applicable calibration whenever the camera position, placement plane, or camera group changes.	**Configure next** After setting detection areas and parameters, always select [Save and send].
Verify afterward Use a representative reference object to verify dimensions, repeatability, result time, and communication.	**Stop on any issue** Do not use the current result if the status is abnormal, calibration does not meet the requirement, or results are unstable.

2 Before You Start

Use this chapter to decide which conditions are directly supported and which may cause detection failure, measurement failure, or inaccurate results.

2.1 Recommended Operating Conditions

Installation The cameras must be rigidly mounted and must not move or vibrate after installation. The maximum supported installation height is 4 m.	**Placement** Place the object horizontally and keep it stationary on a floor or platform that has completed ground calibration.
Size Objects approximately 20 cm or larger are recommended. Keep the complete object inside the camera view and detection area.	**Point cloud** The object surface should produce a continuous depth point cloud without obvious occlusion or large missing areas.



Length, width, height, and volume definitions

2.2 Conditions That May Prevent Detection or Measurement

Black or highly reflective surfaces The camera may not obtain a stable depth point cloud, so the object may not be detected or measured.	**Size and structure** Objects smaller than approximately 20 cm, or objects with thin supports, handles, openings, or missing top surfaces, may be missed or measured with distorted dimensions.
Occlusion and pose Measurements may be inaccurate if the contour or bottom is occluded, or if the object is tilted, suspended, or inverted.	**Shape** [Real Volume] is only an estimate for soft packages, bags, bulk materials, and objects wider at the top.
Multiple objects and environment Touching or closely spaced objects, overlapping areas, strong light, dust, vibration, or measurement during motion may cause detection failure.	**Intended use** Do not use this algorithm for millimeter-level precision measurement, billing settlement, or material net-volume accounting.

Rain and snow The camera is not waterproof. Do not expose it directly to rain, snow or water spray. Use a dry, sheltered environment; precipitation may also degrade depth data. Validate actual samples and the final installation before use.

2.3 Result Definitions and Acceptance Criteria

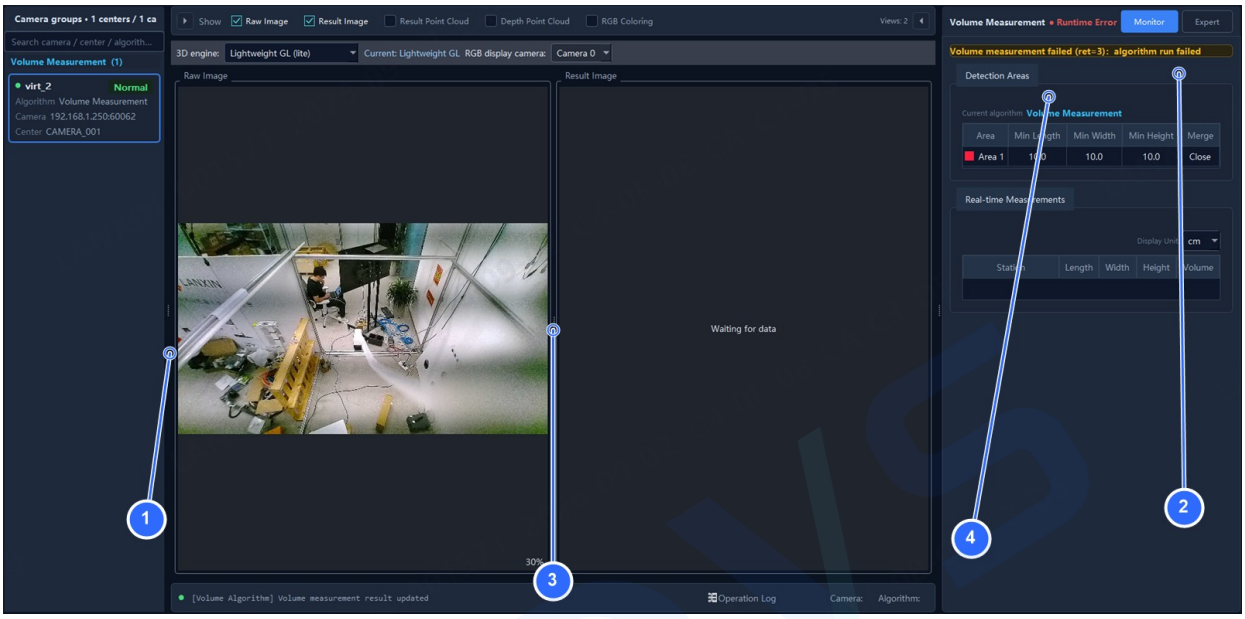
Empty area With no object present, the result should remain empty; background or floating ground points must not be detected as objects.	**Reference object** The object should be detected at both the center and edges, and the dimensional error must meet the project requirement.
Multiple objects The object count and spacing should match expectations. Verify [Merge Area Objects] if the project uses it.	**Continuous measurement** Area IDs must be correct, with no unexpected duplicates. Deviation and result time must meet the project requirement.

Completion check Revalidate after any change to camera position, support plane, object type, software, or parameters. One screenshot or one successful run is not acceptance.

3 Complete the First Measurement

3.1 Main Interface

After selecting a camera group, the main interface consists of the camera-group list on the left, the center view area, and the algorithm panel on the right; the upper-right area shows algorithm status.



Main interface orientation

① Left: camera-group list Use the camera-group list to select the group to configure or run and to check camera connection and algorithm status. Select the target group before any operation.

② Right: mode	Purpose
Monitor	Used for daily operation to check algorithm status, detection areas, and real-time measurement results.
Expert	Used for first-time configuration and debugging, including stitching, ground calibration, detection areas, parameters, and logs.
③ Center: view controls	**Displayed content**
Raw Image	Original color image from the camera
Result Image	Image with algorithm results overlaid
Result Point Cloud	Point cloud produced by the algorithm
Depth Point Cloud	Raw depth point cloud captured by the camera
RGB Coloring	Maps color images onto point clouds; display only

③ Center: viewing results Open [Depth Point Cloud] and [Result Point Cloud] together. Changing the view does not change algorithm parameters or measurement results.

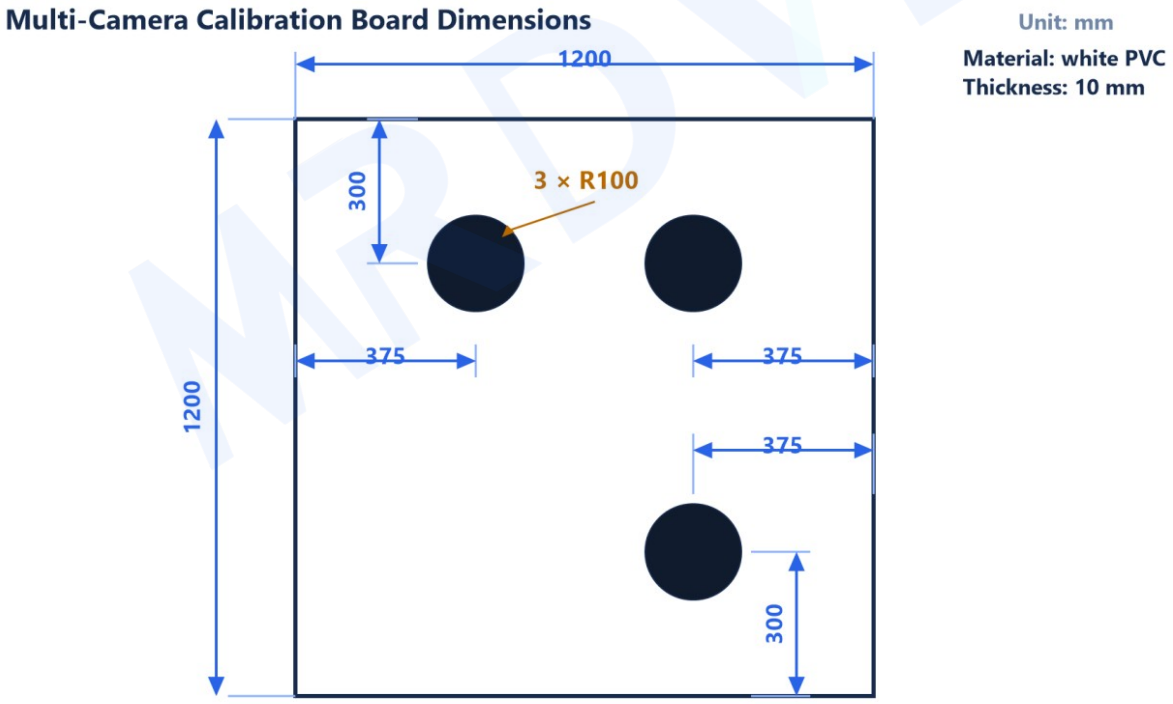
④ Upper right: algorithm status [Runtime Error] may indicate an offline camera, invalid authorization, a missing or stopped AW3 backend, a missing center, unsent parameters, or an algorithm failure. During first setup, it may remain until Sections 3.4-3.5 are complete and [Save and send] succeeds.

3.2 Multi-Camera Calibration

Single-camera groups skip this section If only one camera is configured, continue directly to 3.3 [Ground Calibration].

Calibration requirement Lock every camera in its final position before calibration. Any later movement requires multi-camera calibration and ground calibration again.

3.2.1 Calibration Board Pattern and Size

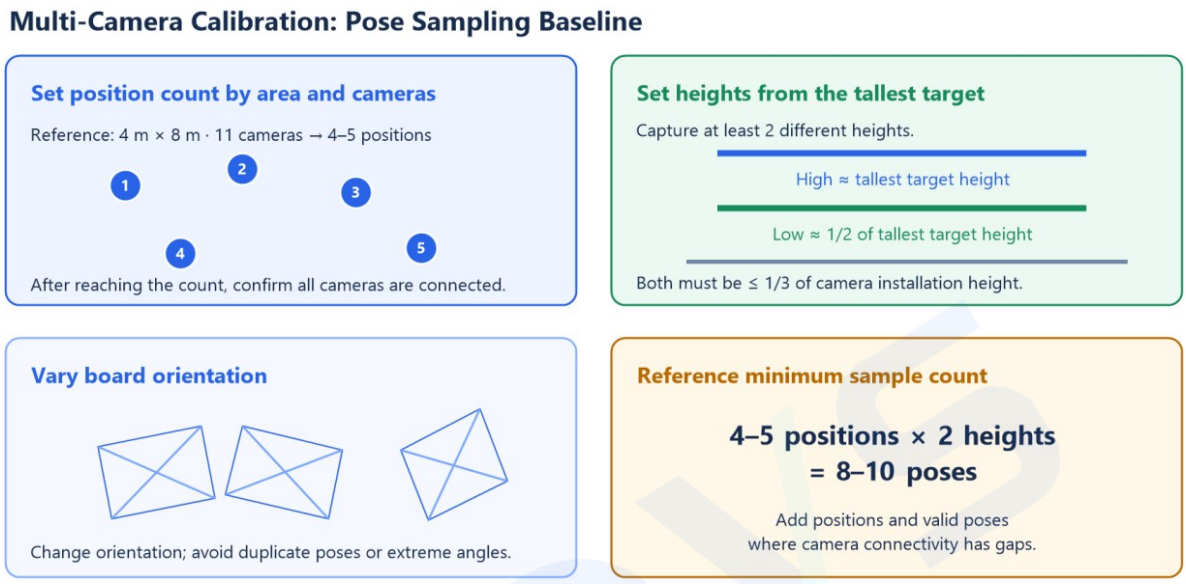


1200 mm × 1200 mm three-circle calibration board

Dimensions The board is 1200 mm × 1200 mm. All three circular markers have a radius of 100 mm. Manufacture the marker positions using the 375 mm and 300 mm dimensions shown in the drawing.

Board requirements Use a 10 mm thick white PVC board. The detection surface must be flat and clean. The black circular markers must be complete and clearly contrast with the white surface. Do not use a warped, damaged, or contaminated board.

3.2.2 Capture Count and Distribution



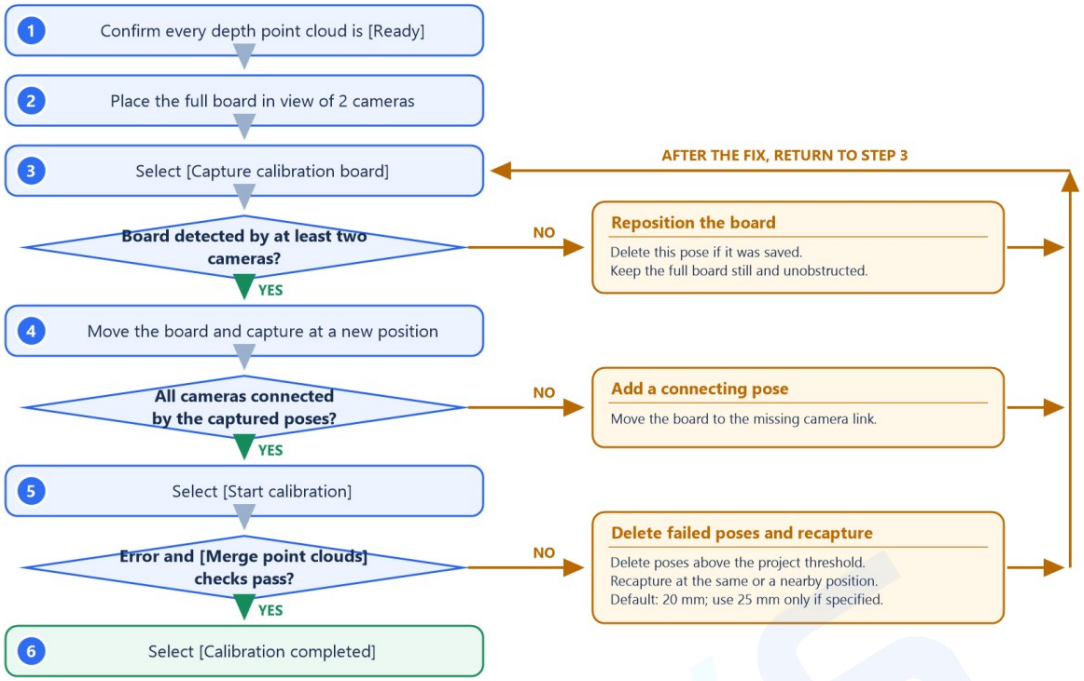
Capture reference for a 4 m × 8 m area with 11 cameras

Position reference For a 4 m × 8 m area with 11 cameras, start with 4-5 positions, about 6.4-8 m² per position. Add positions for larger areas, more cameras, or missing links between adjacent cameras.

Height reference Measure the board height at its center. Capture at two heights: near the tallest target and near half that height. Keep both at or below one-third of the camera installation height; if necessary, lower both while maintaining clear separation.

Completion condition Capture two heights at each of the 4-5 positions, for 8-10 valid poses. Then verify connectivity, per-pose error, and point-cloud overlap; add poses where coverage is missing.

3.2.3 Calibration Workflow



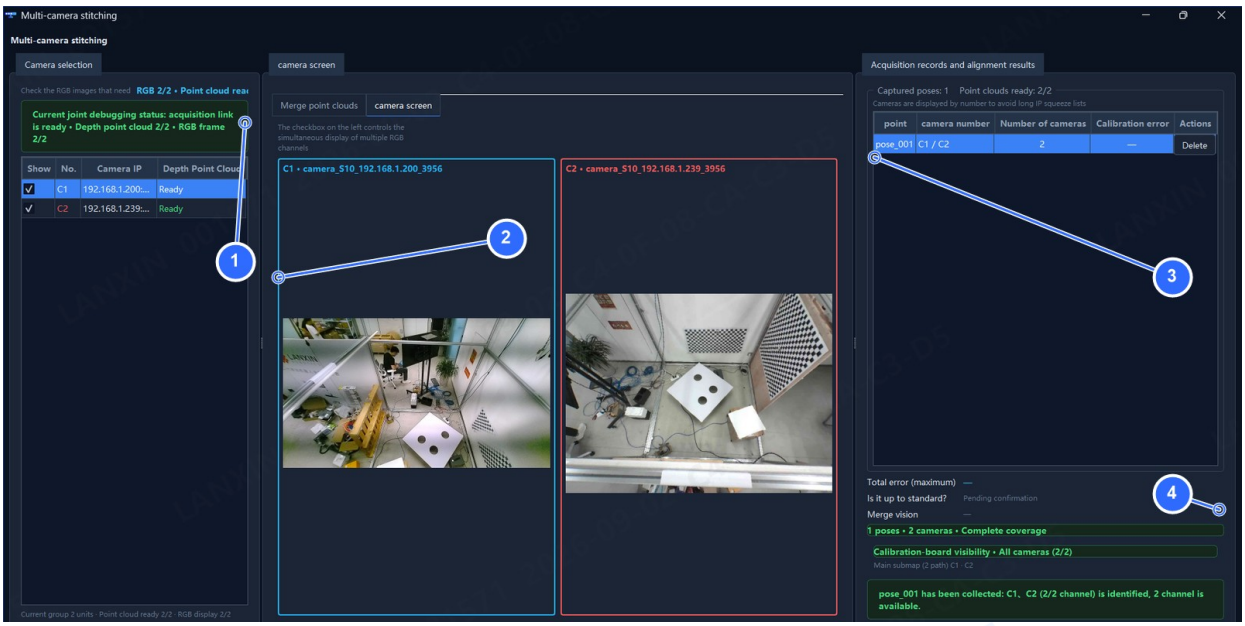
Multi-camera stitching workflow and recovery paths

Connectivity concept The board does not need to be visible to all cameras at once. Use multiple poses to connect all cameras into one calibration group.

3.2.3.1 Capture the First Valid Pose

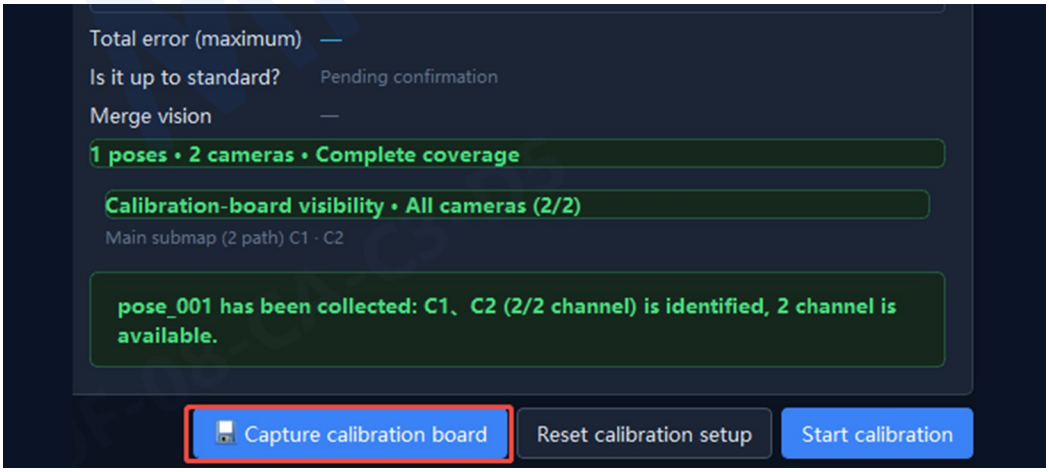
Path: [Expert] > [Calibration] > [Multi-Camera Calibration].

1. Confirm that the depth point cloud for every camera is [Ready].
2. Place the board where adjacent cameras can both see it. Keep the complete board stable and unobstructed.
3. Select [Capture calibration board]. Each capture creates one pose.



Multi-camera calibration: check cameras, place the board, and capture a pose

Page area	Purpose
① Left: cameras and point-cloud status	Select the cameras included in calibration. Every depth point cloud must show [Ready].
② Center: camera views	Confirm that the complete board is stable, unobstructed, and visible in adjacent cameras.
③ Right: acquisition records and alignment results	Check the cameras included in each pose, together with coverage, connectivity, and calibration error.
④ Lower right: total error, coverage, and connectivity	Check total error, calibration-board visibility, camera coverage, and connectivity.

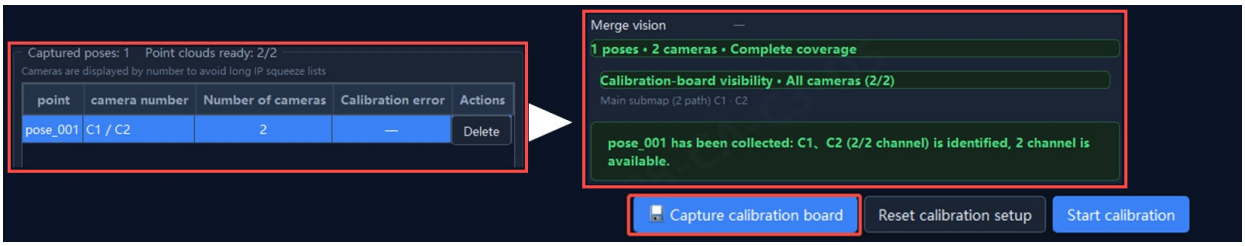


Close-up: acquisition status and controls; each pose should include at least two cameras

3.2.3.2 Add Poses Until All Cameras Are Connected

1. Move or rotate the calibration board and capture it at a new position.
2. At least two cameras must detect the board in each pose.

3. Repeat until the right panel shows [Complete coverage] and all calibration-board paths are connected.
4. Select [Start calibration].

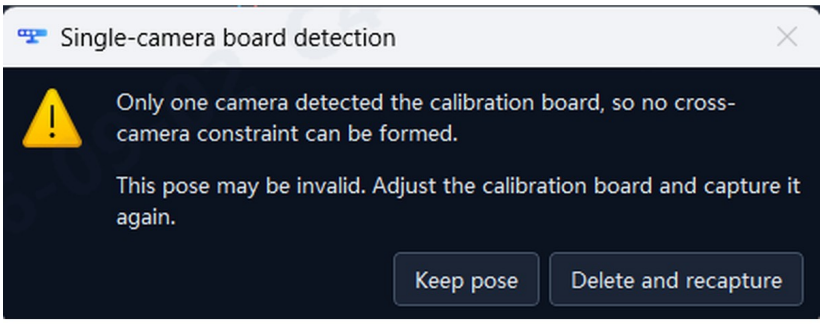


Start calibration after acquisition records are complete

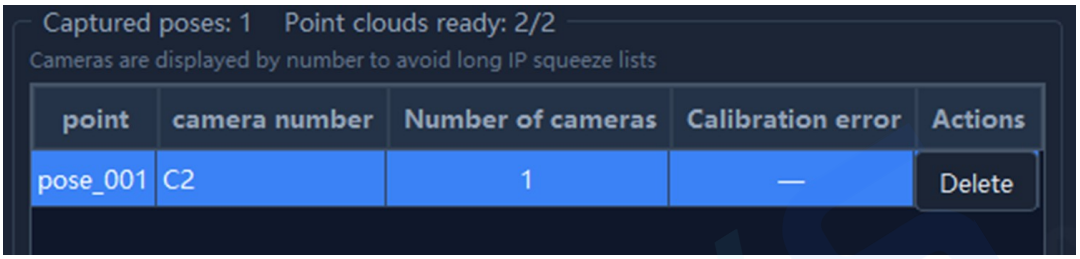
Check item	Acceptance criterion
Acquisition records	Each pose and the cameras included in that pose.
Status panel	Select [Start calibration] only after complete coverage and connectivity are confirmed.

Not connected If all cameras never become connected, follow the missing-link prompt and place the board inside the overlapping view of the indicated camera pair. Do not repeatedly capture at the same position.

What if only one camera is detected?



Capture warning: only one camera detected the board in the current pose



Pose record: check participating cameras and camera count

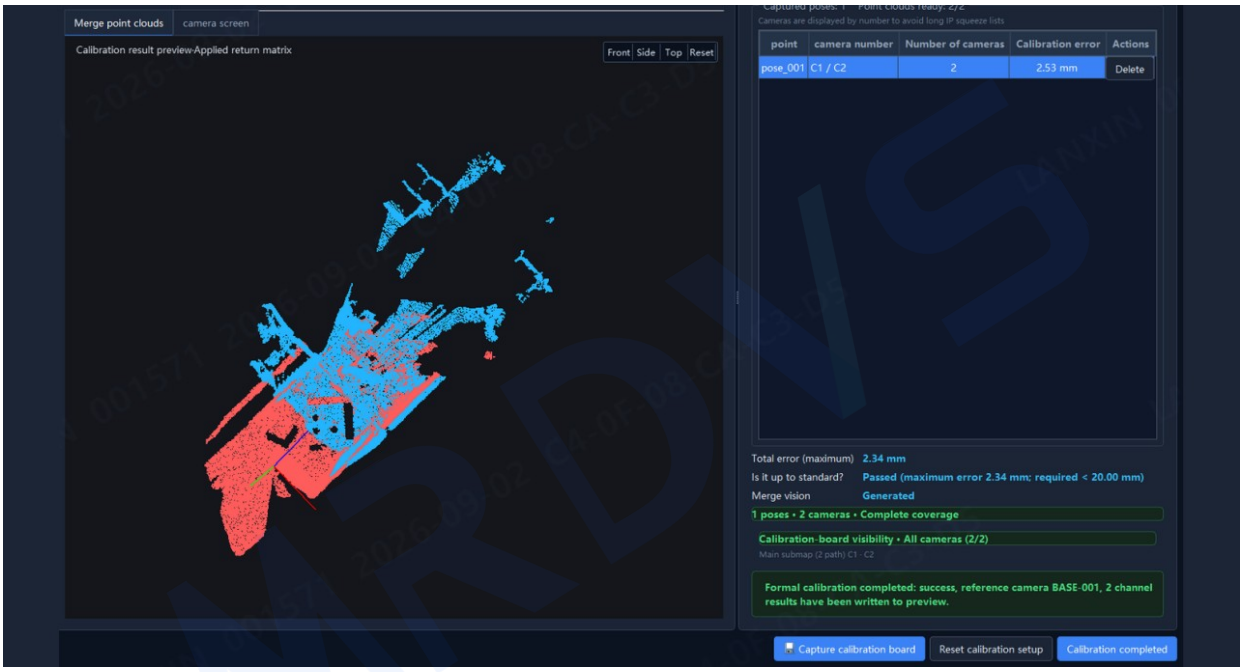
1. Select [Delete and recapture].
2. Move the board into the overlapping view of adjacent cameras and adjust its orientation to increase the visible area.
3. Confirm that both depth point clouds are ready, and avoid occlusion, reflection, and rapid movement.

Viewing angle If the board angle is too steep, the circles may distort or disappear and capture may fail. Reorient the board until adjacent cameras clearly see all three circles, then recapture.

Stop condition Do not use [Keep pose] unless project debugging explicitly requires it. A pose detected by only one camera cannot establish a camera connection.

3.2.3.3 Check and Complete Calibration

1. Select [Merge point clouds] and check whether the same surface from different cameras nearly overlaps without obvious ghosting or offset. This preview is different from [Merge Area Objects].
2. Check [Total error (maximum)] and [Is it up to standard?]. The default threshold is 20 mm; use 25 mm only when the project explicitly accepts it.
3. If the total error exceeds the current project threshold, check each [Calibration error]. Delete poses that exceed the same threshold, recapture the board at the original or nearby position, and select [Start calibration] again.
4. Confirm that coverage and calibration-board connectivity are both green.
5. Select [Calibration completed] to return to the AW3 main interface.



Merge point clouds, pose error, total error, calibration status, and action buttons

Check item	Acceptance criterion
Merge point clouds	Overlays the participating camera point clouds to check overlap. The same surface should show no obvious ghosting or offset. This is not [Merge Area Objects].
Total error / standard status	Evaluate the overall result against the project threshold: 20 mm by default, or 25 mm when explicitly accepted by the project.
Per-pose calibration error	If the total error exceeds the threshold, check each pose, delete poses above the same threshold, and recapture at the original or a nearby position.
Reset / complete calibration	Select [Reset calibration setup] to change participating cameras or restart the current calibration, then capture and calibrate again. Select [Calibration completed] only after error, coverage, and connectivity meet the requirement.

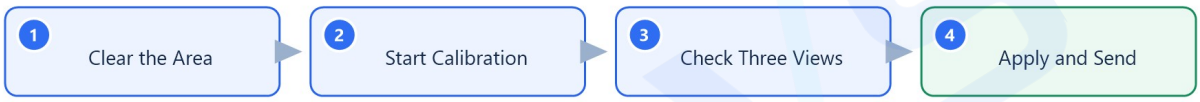
Error meaning Calibration error measures stitching quality, not dimensional accuracy. Meeting the threshold does not guarantee length, width, or height accuracy; validate tighter requirements separately.

Stop condition Do not complete calibration if the error exceeds the project threshold or the point clouds are visibly misaligned. Delete high-error poses, recapture the affected area, and recalibrate.

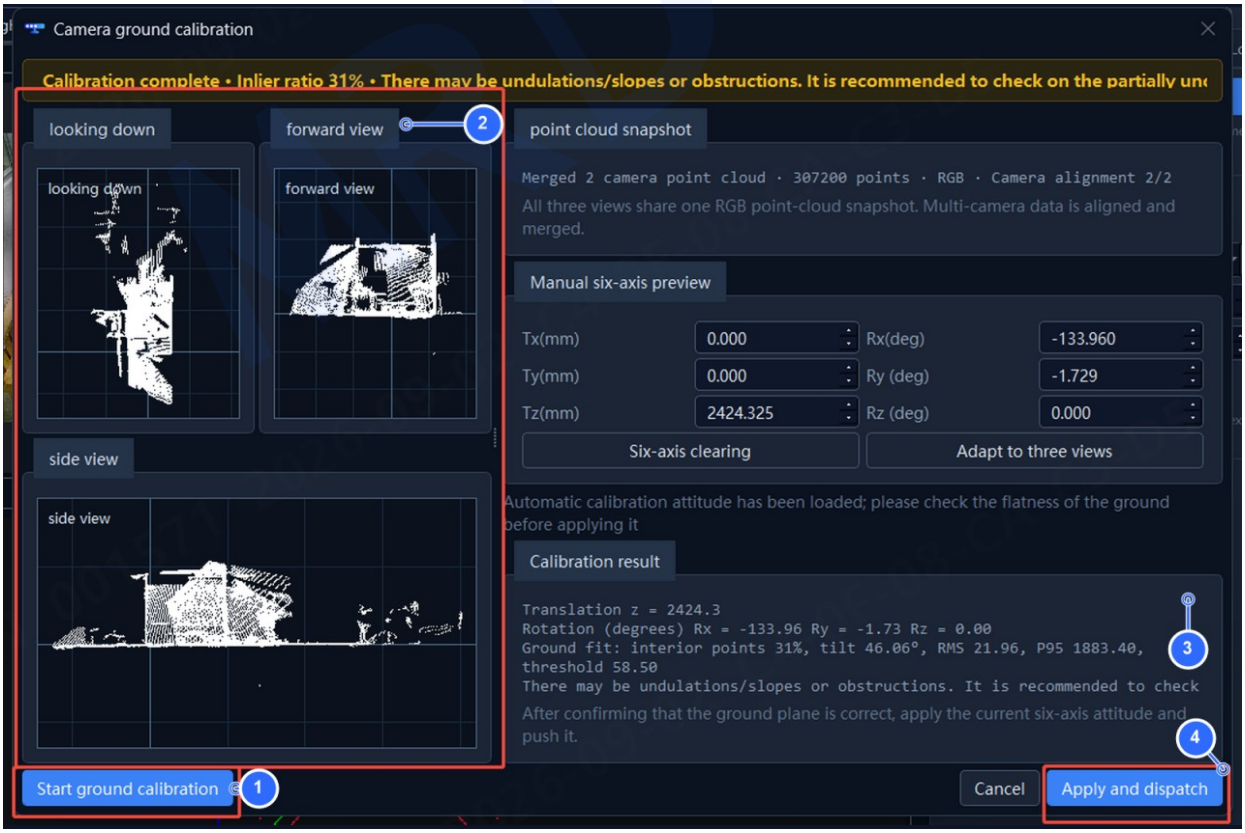
3.3 Ground Calibration

Ground calibration defines the floor or platform on which objects are placed and establishes the measurement zero plane. It is required for both single-camera and multi-camera groups. Complete multi-camera stitching first when applicable.

Calibration requirement Remove every object from the camera view so the floor or platform is fully exposed. The plane must be flat and unobstructed.

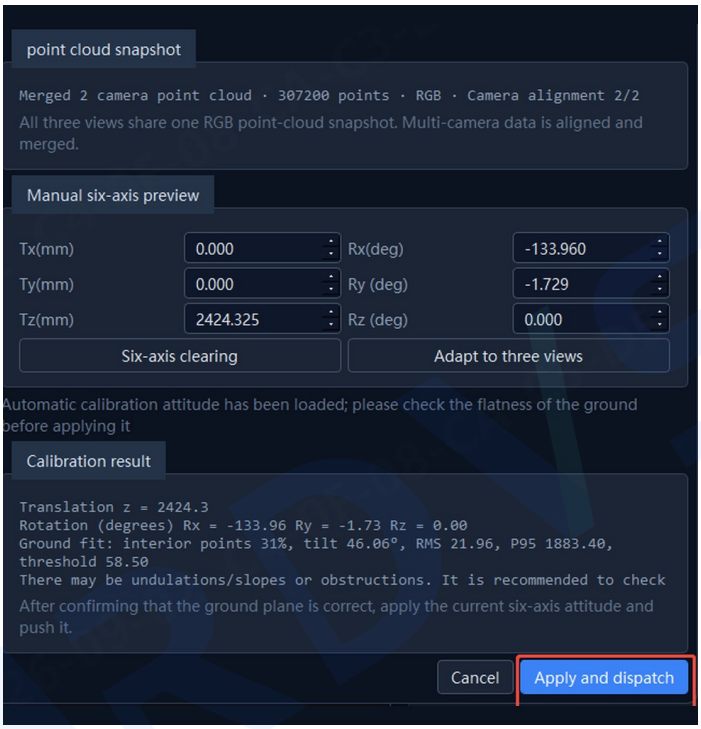


Ground calibration in four steps



Ground calibration window

Page area	Purpose
① Start ground calibration	Automatically detects the floor or platform.
② Looking-down, forward, and side views	Check whether the plane is nearly level and whether any object or support is included.
③ Warnings and calibration result	Check the inlier ratio, tilt, and fit result. If a warning appears, correct the scene and calibrate again.
④ Apply and dispatch	After verification, save and send the ground reference.



Close-up: point-cloud snapshot, six-axis preview, calibration result, and dispatch button

Stop condition If the result shows a low inlier ratio, uneven ground, slope, or obstruction, clear the area, verify the plane, and recalibrate. Do this even if the window says calibration is complete.

Completion check The placement plane is nearly level in all three views, the zero plane matches the actual surface, and dispatch succeeds.

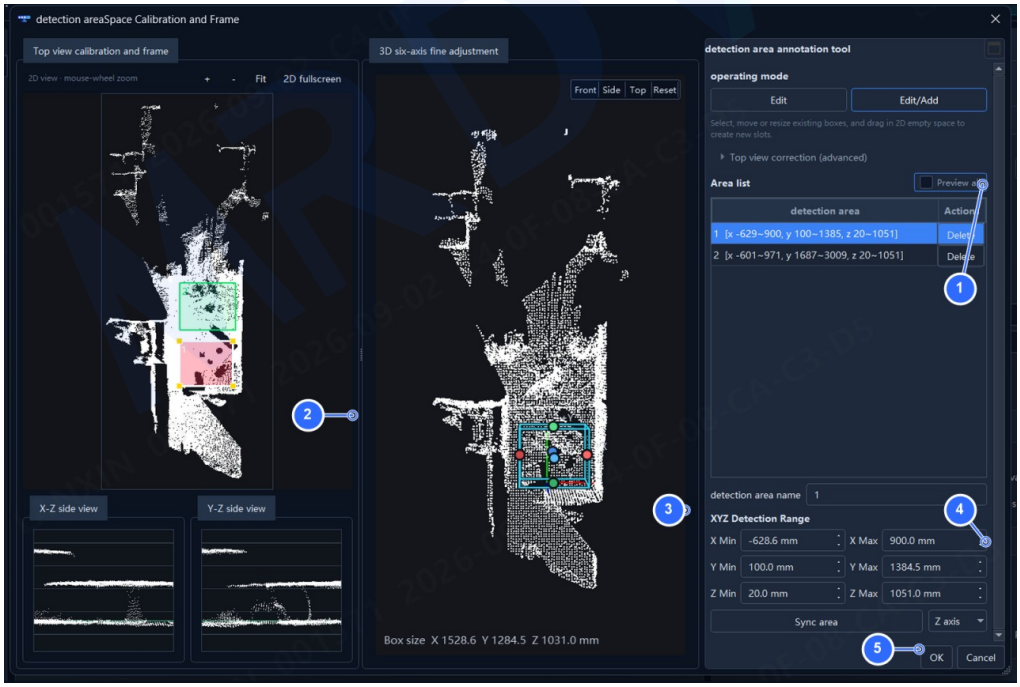
3.4 Set Detection Areas

Path: [Expert] > [Volume measurement] > [Add/Edit Areas]. Complete multi-camera stitching, if applicable, and ground calibration first.



Detection-area setup workflow

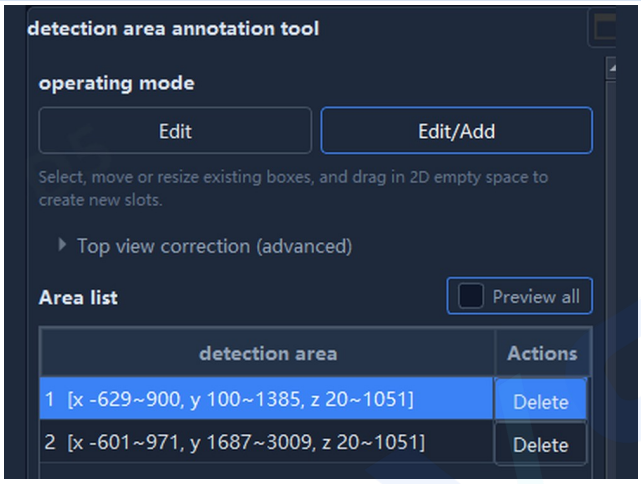
1. Select an operating mode: [Edit] adjusts only the selected existing area; [Edit/Add] can adjust existing areas and create a new one by dragging in empty space in the 2D top view.
2. Move the area in the 2D top view, or drag its corners to adjust the X-Y position and range.
3. Drag the handles in the 3D view to adjust the 3D range, and use the X-Z and Y-Z views to check height and boundaries.
4. Check the area name and the X, Y, and Z minimum and maximum values on the right. Exclude supports, background, and unrelated objects.
5. Select [OK] to return to the main page, then continue with the detection parameters.



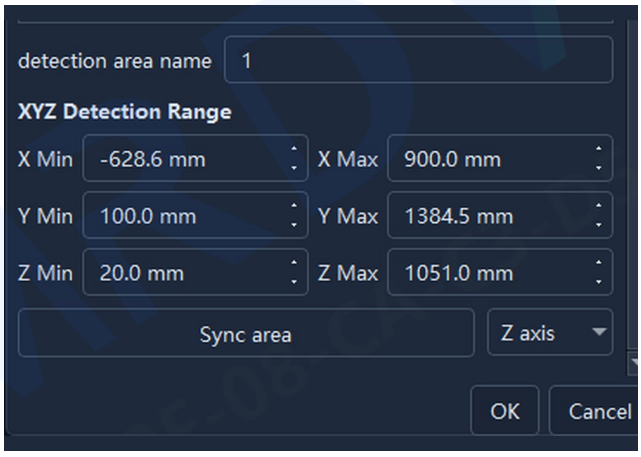
Detection-area calibration and box editing

Page area	Purpose
① Upper right: operating mode and area list	Select [Edit] or [Edit/Add], then select, delete, or preview areas.
② Left: 2D top view	Create an area, move it, and adjust its X-Y position and range.

Page area	Purpose
③ Center: 3D view	Drag the handles to adjust the 3D range; use the X-Z and Y-Z views to check height.
④ Lower right: XYZ range	Precisely check each axis minimum, maximum, and the area name.
⑤ Lower right: OK	Saves the area edit and returns to the main page; it does not send algorithm parameters.



Close-up: Edit and Edit/Add modes



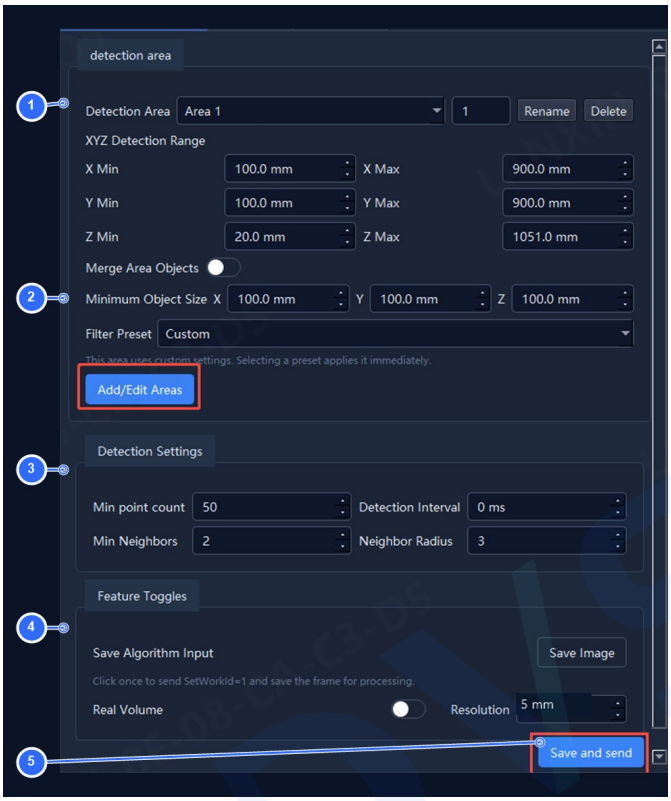
Close-up: XYZ range and OK button

Z-axis reference Z=0 is the calibrated floor or platform. For a raised support or pallet, set [Z Min] to its top surface. Too low includes the support; too high cuts off the object bottom.

Do not skip [OK] only saves the area edit; it does not send parameters. After Section 3.5, always select [Save and send].

3.5 Set Detection Parameters

Path: [Expert] > [Volume measurement]. Configure each detection area separately. For initial use, keep the project-delivered values and do not copy values from screenshots.



Detection area, filter preset, feature toggles, and save entry

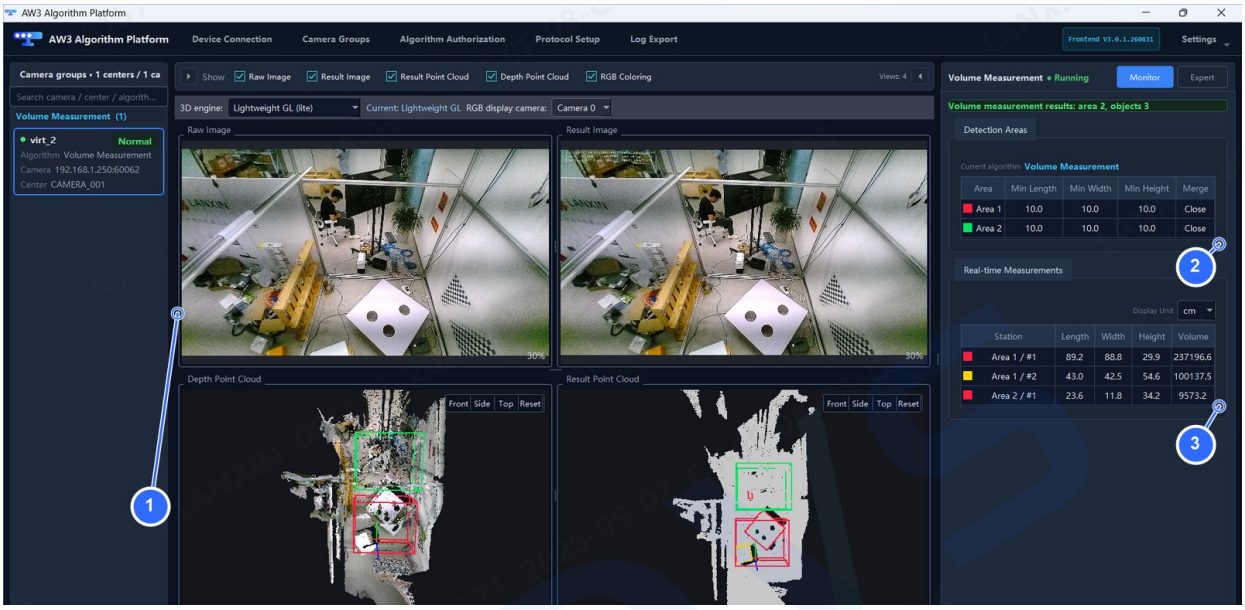
Page area	Purpose
① Detection Area	Select the current area and check the XYZ range and Minimum Object Size.
② Filter Preset	Select one of the four project presets. See Section 4.2 for applicable scenarios.
③ Detection Settings	Set Min point count and the neighborhood-filter thresholds.
④ Feature Toggles	Enable [Save Image], [Merge Area Objects], or [Real Volume] only when needed.
⑤ Save and send	Sends the current area and parameters to the algorithm.

1. Select the detection area and set [Minimum Object Size X / Y / Z]. This parameter is a filter; it does not define minimum measurable size or accuracy.
2. Select one of the four project presets according to the main surface orientation. See Section 4.2.
3. Check [Min point count], [Min Neighbors], and [Neighbor Radius]. Do not change them manually without a clear debugging basis.
4. Keep [Merge Area Objects] and [Real Volume] off during initial verification. Enable them only after basic measurement is stable and after reviewing Chapter 4.
5. Select [Save and send] and confirm that the parameters were sent successfully.

Completion check [Runtime Error] no longer appears, the area is correct, and the reference object produces stable results.

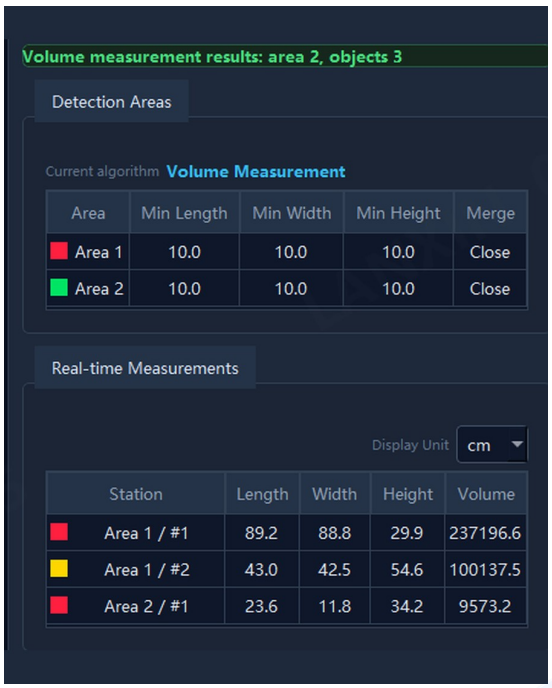
3.6 Monitor and Verify Measurement Results

After detection areas and parameters are sent, switch to [Monitor]. Understand the page and color mapping before verifying with a reference object.



Monitor page overview

Page area	Purpose
① Center view area	Compare [Raw Image] with [Result Image], and [Depth Point Cloud] with [Result Point Cloud]. Select the camera in [Raw Image]. Keep both point-cloud views open for 3D checks.
② Detection Areas table	Shows area colors, minimum sizes, and the [Merge Area Objects] status.
③ Real-time Measurements table	Shows the length, width, height, and volume of every detected object in each area.

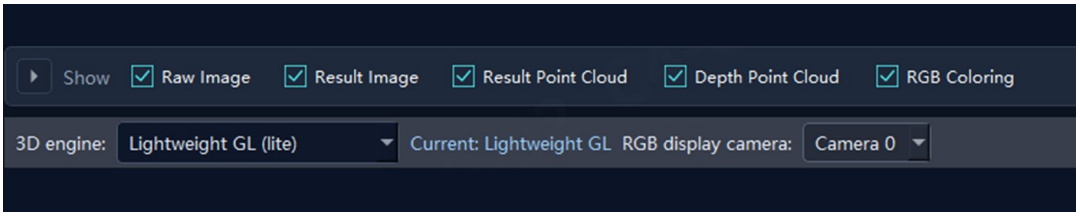


Close-up: Detection Areas and Real-time Measurements tables

3.6.1 Understand the Two Tables and Colors

Item	Description
Detection Areas table	One row per area. The color matches Result Point Cloud; Merge shows object merging.
Real-time Measurements table	One row per object. "Area 2 / #1" means object 1 in Area 2.
Area color	Matches the area box color. Up to five areas.
Object color	Matches the object box color. Up to ten colors per area; colors repeat between areas.

3.6.2 Switch Views and Apply RGB Coloring



Close-up: View switches

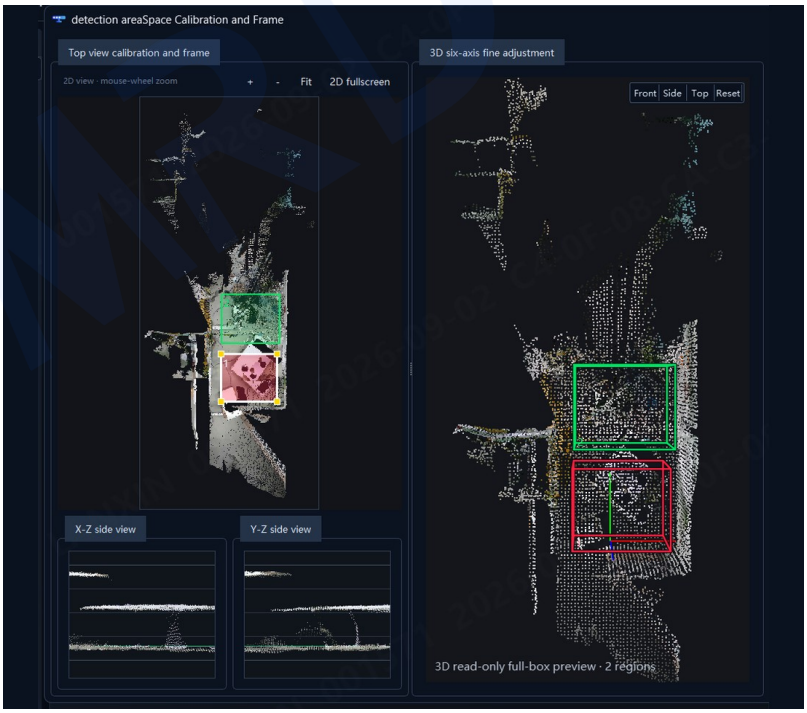
Function	Description
RGB camera selection	[RGB display camera] selects which camera [Raw Image] shows.
Result Image	Shown automatically after detection; the camera cannot be selected here.
Point-cloud view	Changes the view angle only; coordinates, areas, and results stay unchanged.

Function	Description
RGB Coloring	Adds RGB color to the point-cloud display only.

Viewing results Open [Depth Point Cloud] and [Result Point Cloud] together.



Depth and Result Point Clouds with RGB Coloring



Detection-area editor with RGB Coloring

3.6.3 Verify with a Reference Object

1. Check status, both images, both point clouds, and the results table.
2. Measure a known object; check count, contour, L/W/H, and volume.

3. Compare the reference and result in the same unit.
4. Repeat at several positions; confirm accuracy, repeatability, and result time.

Screenshot note [Not Authorized] only records the captured UI state; it is not acceptance evidence.

3.6.4 Verify Four Scenarios

Empty area No object; background and floor are not detected.	**Single reference object** Test center, edge, and camera overlap; dimensions meet the project limit.
Multiple objects Count and spacing are correct; merged output matches the setting.	**Continuous and multi-area operation** Area IDs, duplicates, and result time meet the project requirement.

Completion check Accuracy, repeatability, and result time all pass.

3.7 Start Daily Measurement



Start-up sequence for every power-on

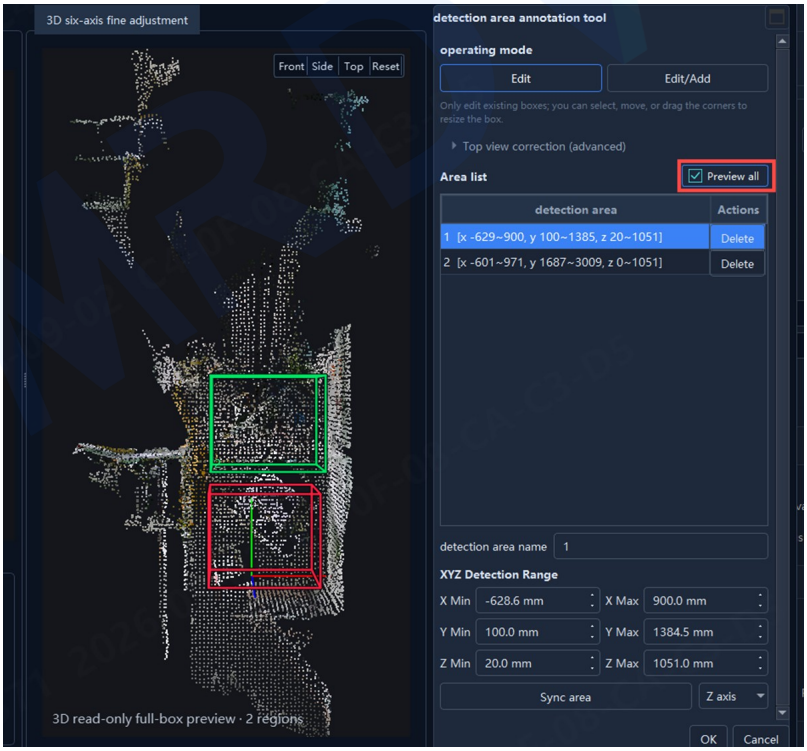
Operating rule If any status or measurement error appears, stop using the result and follow Chapter 6.

4 Functions and Parameters

Use this chapter after the main workflow is complete. Finish Chapter 3 before adjusting these functions.

4.1 Manage Detection Areas

Function	Description
Edit	Adjusts only the selected existing area. Move it or drag its corners to change the range.
Edit/Add	Adjusts existing areas and creates a new area by dragging in empty space in the 2D top view.
Preview all	Displays all areas together. The 3D view is read-only in this mode.
Sync area	Copies the selected axis minimum and maximum to all selected areas.
Rename / Delete	Changes the displayed name or removes an area. Before deletion, check the area ID and its mapping in the receiving system.
OK	Saves the area edit and returns to the main page. [Save and send] is still required.



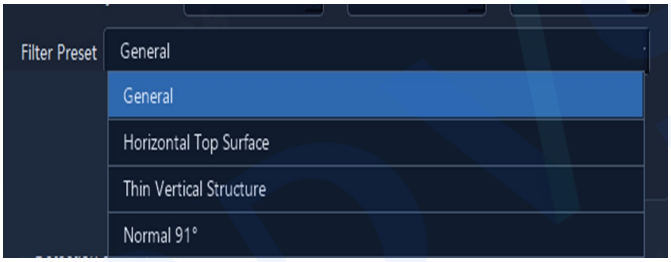
Close-up: Preview all and 3D area boxes

Areas and coordinates Overlapping areas can report the same object more than once. X/Y set the horizontal range; Z sets height. See Section 3.4 for Z Min.

4.2 Adjust Detection and Filtering Parameters

Parameter (Default)	Purpose and effect
Min point count (50)	Filters point-cloud clusters with too few points. A high value may miss objects; a low value may retain noise.
Detection Interval (0 ms)	Sets the interval between detections. It is not Hz and does not equal end-to-end result latency.
Min Neighbors (2)	Filters noise by nearby point count. Excessive filtering may damage the object contour.
Neighbor Radius (3 mm)	Defines the distance considered nearby and works together with Min Neighbors.
Real Volume Resolution (5 mm)	Used only with [Real Volume]. It sets the estimation grid, not measurement accuracy. Smaller values require more processing.

Filter Presets and Applicable Scenarios



Filter-preset reference

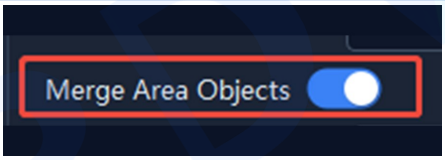
Project preset	Applicable scenario
General	Use for typical surface orientations or when the starting preset is not yet known. It is the initial verification baseline.
Horizontal Top Surface	Use when the object top is approximately horizontal and measurement mainly depends on that top surface; its normal is close to vertical.
Thin Vertical Structure	Use for plates, sheets, or thin structures dominated by vertical surfaces; their effective-surface normals are close to horizontal.
Normal 91°	Use for curved or inclined surfaces, or surfaces with widely varying orientation, when other presets remove valid surfaces and cause contour loss.

4.3 Merge Area Objects and Real Volume

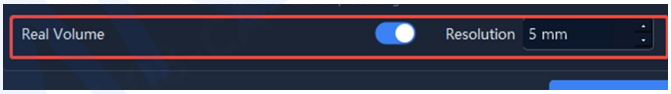


When [Real Volume] is on, the table shows bounding-box volume and a Real Volume estimate.

Function	Description
Merge Area Objects	Outputs multiple objects in the same area as one combined object, including the gaps between them. It is not a sum of individual volumes and is not multi-camera stitching.
Real Volume	Real Volume uses surfaces visible to the cameras. It is an estimate for reference only, not internal capacity or net material volume. Default [Resolution]: 5 mm.



Close-up: Merge Area Objects switch

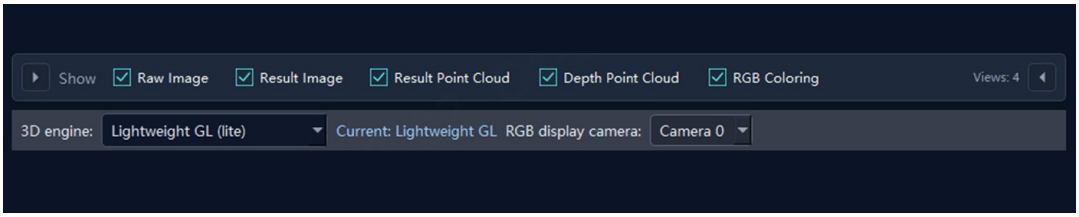


Close-up: Real Volume switch and Resolution

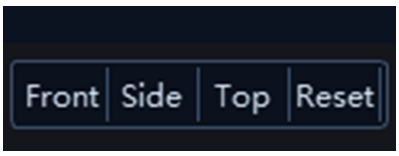
Parameter meaning [Resolution] sets the estimation grid in millimeters, not accuracy. Smaller values require more processing.

Limitations [Real Volume] is an estimate for reference only and must not be used for billing. It may read high for soft, loose, wider-at-the-top, suspended, inverted, or bottom-occluded objects. See Chapter 2.

4.4 RGB Camera, Point-Cloud Views, and Save Image

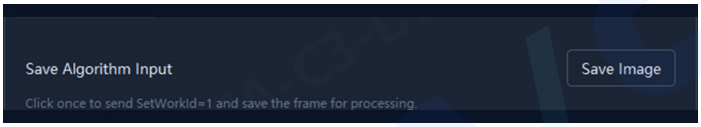


Close-up: RGB camera and image-display controls



Close-up: Front, Side, Top, and Reset point-cloud buttons

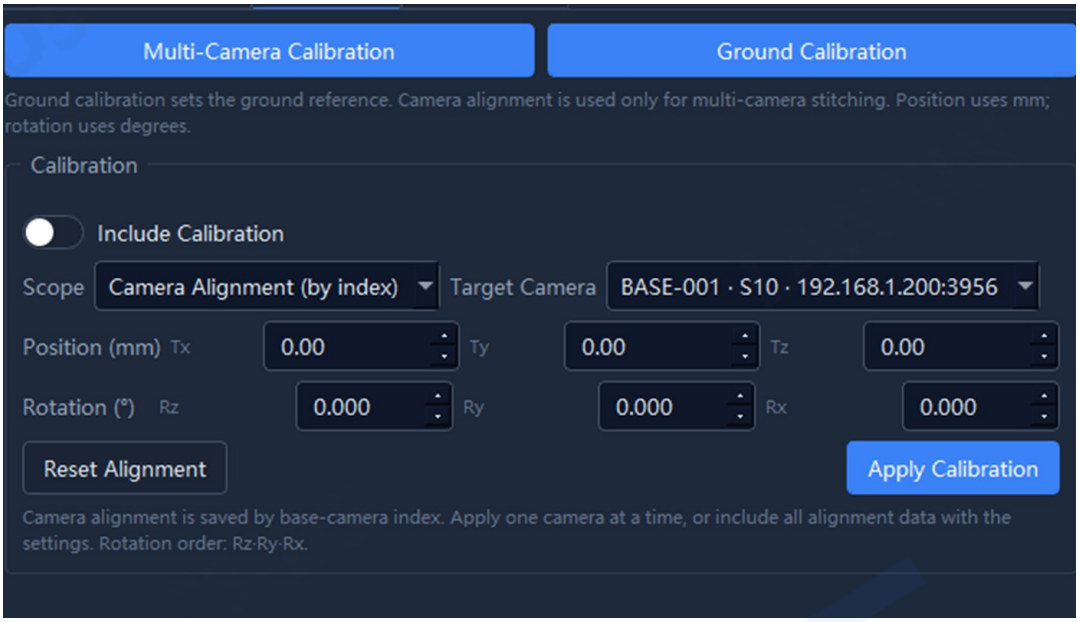
Function	Operation and effect
RGB camera selection	Select the preview source under [RGB display camera]. [Raw Image] displays the selected camera.
Point-cloud view	Select [Front] / [Side] / [Top] / [Reset]. This changes only the viewing angle, not coordinates, areas, or results.
Result Image	Displays detection results automatically and cannot switch cameras. Use [Raw Image] when another camera view is needed.
RGB Coloring	Maps RGB colors onto the point cloud and changes display only. See Section 3.6.2.



Close-up: Save Image button

1. Open [Expert] > [Volume measurement] and locate [Save Algorithm Input] under [Feature Toggles].
2. Select [Save Image] when reproduction data is required, then wait for the algorithm to finish processing.
3. Click [Log Export] on the top bar, select Volume Measurement and the required data categories, then verify and export the files.

4.5 Calibration Parameters



Calibration settings: generate parameters through the calibration workflows whenever possible

Parameter	Purpose
Camera alignment	Relative camera positions used to align multi-camera point clouds.
Global alignment	The camera group's coordinate reference relative to the ground, generated by ground calibration.

Warning [Reset Alignment] zeros all translation and rotation values. It does not restore the previous calibration and will change measurement coordinates.

5 Communication and Result Data

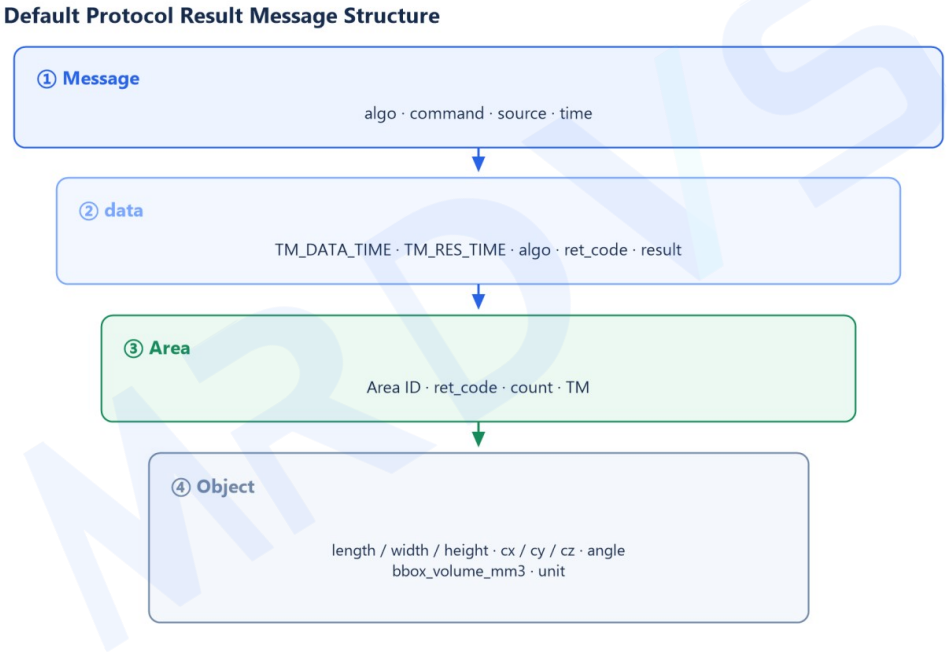
Supported protocol methods TCP (persistent or short-lived), UDP, and HTTP, with default JSON messages. Push (active sending) and Serve (listening) modes are available; configure trigger input and result forwarding in AW3.

5.1 Configure and Verify Communication

Configure the operating mode, trigger input, and result forwarding in AW3 as described in Section 7.4 of the AW3 User Manual.

Integration acceptance Complete a measurement in the selected mode and check that the receiving system gets the Section 5.2 result. If external triggering is used, verify the trigger request too.

5.2 Read Result Messages



Default protocol: navigate from the message to data, area, and object fields

Field	How to read it
data.ret_code	Check the frame status first. If it is non-zero, the frame result is invalid.
data.result.<ID>.ret_code	Then check the area status. If it is non-zero, that area result is invalid.
count / TM	count is the object count; TM is the object array. If both frame and area ret_code are 0, count=0 with TM=[] means a valid empty area.
Units	length, width, height, cx, cy, and cz use mm; angle uses degrees; bbox_volume_mm3 uses mm³.
Time	data.TM_DATA_TIME is the acquisition time and data.TM_RES_TIME is the result time. Measure end-to-end latency separately.

Default Protocol JSON Example (Illustrative Data)


```
{
  "algo": "TorusMetric",
  "command": "RESULT_UPDATE",
  "data": {
    "TM_DATA_TIME": "2026_09_01_15_59_37.017",
    "TM_RES_TIME": "2026_09_01_15_59_37.346",
    "algo": "torusmetric",
    "result": {
      "1": {
        "TM": [
          {
            "angle": 63.561988830566406,
            "bbox_volume_mm3": 114474240,
            "cx": 499, "cy": 450, "cz": 447,
            "height": 855, "length": 1046,
            "unit": "mm", "width": 128
          },
          {
            "angle": -27.813127517700195,
            "bbox_volume_mm3": 55437770,
            "cx": -197, "cy": 605, "cz": 217,
            "height": 394, "length": 1315,
            "unit": "mm", "width": 107
          },
          {
            "angle": -27.776199340820313,
            "bbox_volume_mm3": 272973870,
            "cx": 42, "cy": -11, "cz": 527,
            "height": 1014, "length": 1965,
            "unit": "mm", "width": 137
          }
        ],
        "count": 3, "ret_code": 0
      }
    },
    "ret_code": 0
  },
  "source": "vc_3905aff4-3c80-412b-934a-968c708ac280",
  "time": "2026-09-01 15:59:37"
}
```


Version note For the default protocol, use the hierarchy shown here; do not use the obsolete top-level IP/TM format. Follow the project interface document for error codes.

6 Troubleshooting and Maintenance

First determine whether Raw Image and Depth Point Cloud continue to update, then follow the appropriate branch.

Observation	Go to
No image, frozen image, or offline camera	6.1 System Issues (Streaming)
Streaming is normal, but results are missing, inaccurate, or grouped incorrectly	6.2 Algorithm Performance Issues

Stop using results first During an issue, do not use a stale image, previous measurement value, or unconfirmed external result.

6.1 System Issues (Streaming)

Symptom	Action
Center not found / no image	Check service IP, port, network, and backend. Start it if stopped; install it if missing (AW3 User Manual).
Camera offline / stream interrupted	Check camera power, network cables, camera-group status, and authorization. After recovery, confirm that Raw Image and Depth Point Cloud update continuously.
Image stops updating	Check the AW3 backend and camera connections. Revalidate after streaming resumes and do not use stale results from before recovery.

6.2 Algorithm Performance Issues

Symptom	Action
[Runtime Error] appears on the right	Complete detection-area and parameter setup and select [Save and send]. If the error remains, check the logs.
Image is available but there is no result	Check whether the object is inside the area and whether size, point-count, or filter parameters remove it.
Dimensional deviation / missing contour	Check point-cloud completeness, multi-camera stitching, ground calibration, area boundaries, material, and occlusion in that order.
Multiple objects merged / duplicate output	Check object spacing, [Merge Area Objects], and overlapping detection areas.
Only one camera detects the calibration board	Delete the invalid pose. Move the board into the overlapping view of adjacent cameras, adjust its orientation, and avoid an excessive viewing angle, occlusion, or reflection.
Stitching offset / calibration does not meet the requirement	Confirm that no camera moved. Check coverage, connectivity, and pose errors, then add captures at missing or high-error connections.

Symptom	Action
Ground-calibration warning	Clear the view so the floor or platform is fully exposed. Remove slopes, uneven areas, or an incorrect plane, then calibrate again.

Communication issue If local measurement is normal but the external system receives no result, use Chapter 5 to check result forwarding, address, port, and the receiving service.

6.3 Export Troubleshooting Data



Log / Export workflow

1. Record the fault time and error. Open [Log Export] on the top bar.
2. Select Volume Measurement, categories, formats and time; click [Batch export ZIP] (AW3 User Manual, Ch. 8).
3. Unzip and check logs, images, and JSON before clearing logs.

Provide one package Versions, camera group, settings, site photos, reference size, abnormal result, and steps.

6.4 Maintenance and Revalidation

Change	Required action
Camera moved	Repeat stitching (multi-camera), ground calibration, and area checks.
Floor height changed	Redo ground calibration; check Z ranges.
Software or setting changed	Back up; recheck accuracy, repeatability, and cycle time.
Routine maintenance	Check lens, mount, and cables; do not move cameras.

Before resuming Back up, then run the four checks in Section 3.6.

Revision History and Technical Support

Version	Date	Revision
V0.1	2026-09-09	Updated 3D illustration and rain/snow restrictions.
V0.2	2026-10-28	Updated protocols and AW3 references; corrected setup, export, troubleshooting, and message descriptions.

Technical Support

Zhejiang MRDVS Technology Co., Ltd.

Hotline: +86 400-025-6680

Website: www.mrdvs.com

GitHub: https://github.com/Lanxin-MRDVS

Knowledge Community: https://docs.mrdvs.com/en

LinkedIn: https://www.linkedin.com/company/mrdvs

Scan to open MRDVS on GitHub